DF Sports Arena - Complete Documentation

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Project Overview

DF Sports Arena is a complete web application for managing a football arena in Adjumani, Uganda. It consists of:

- **Frontend**: Responsive static website (HTML/CSS/JS)
- **Backend**: RESTful API (Node.js/Express)
- **Database**: MongoDB with Mongoose ODM

Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, Bootstrap 5, Vanilla JavaScript
Backend	Node.js, Express.js
Database	MongoDB, Mongoose ODM
Auth	JWT (JSON Web Tokens)
Storage	Cloudinary (images)
Notifications	Twilio (WhatsApp), Nodemailer (Email)

Frontend Documentation

File Structure

```
DF SPORTS ARENA/
├── index.html           # Homepage
├── matches.html         # Upcoming matches
├── flyers.html          # Event flyers
├── gallery.html         # Photo gallery
```

```
├── booking.html         # Booking form
├── contact.html         # Contact page
├── css/
│   └── style.css        # Custom styles (900+ lines)
├── js/
│   ├── script.js        # Main JavaScript
│   ├── api.js           # API service layer
│   └── api-service.js   # API utilities
└── images/              # Local images
```

Pages

1. Homepage (index.html)

- Hero section with animated background
- Upcoming matches display (4 cards)
- Book pitch CTA section
- Google Maps embed
- Footer with navigation

2. Matches Page (matches.html)

- Page header with breadcrumb
- Match cards rendered dynamically
- WhatsApp ticket booking integration

3. Flyers Page (flyers.html)

- Grid layout of event flyers
- Click-to-open modal for preview
- Contact CTA section

4. Gallery Page (gallery.html)

- Responsive photo grid
- Hover overlay effect
- Lightbox modal preview

5. Booking Page (booking.html)

- Full booking form with validation
- Real-time API submission
- Pricing display
- WhatsApp fallback button
- Success confirmation modal

6. Contact Page (contact.html)

- 6 contact cards

- Google Maps embed
- Social media links

Components

Navigation Bar

```
<nav class="navbar navbar-expand-lg navbar-dark fixed-top">
    <!-- Logo -->
    <!-- Mobile hamburger menu -->
    <!-- Navigation links -->
</nav>
```

Match Card

```
<div class="match-card">
    <div class="card-header">
        <h4>Tournament Name</h4>
    </div>
    <div class="card-body">
        <span class="tournament-badge">Upcoming Match</span>
        <div class="teams-display">
            <span class="team-name">Team A</span>
            <span class="vs-badge">VS</span>
            <span class="team-name">Team B</span>
        </div>
        <!-- Date, Time, Booking Button -->
    </div>
</div>
```

Booking Form Fields

Field	Type	Required	Validation
customerName	text	Yes	Max 100 chars
customerPhone	tel	Yes	-
customerEmail	email	No	Valid email format
bookingType	select	Yes	hourly/match/tournament
preferredDate	date	Yes	Future dates only
preferredTime	text	Yes	-
duration	select	Yes	1-6 hours
numberOfPlayers	number	Yes	Min 1

Field	Type	Required	Validation
teamName	text	No	-
opponentTeam	text	No	-
notes	textarea	No	Max 500 chars

API Integration (script.js)

```
// Falls back to static data if API unavailable
async function fetchFromAPI(endpoint, fallback = null) {
    try {
        const response = await fetch(`${API_URL}${endpoint}`);
        const data = await response.json();
        return data.success ? data.data : fallback;
    } catch (error) {
        return fallback; // Use static data
    }
}

// Load functions with API fallback
async function loadMatches() {
    const apiData = await fetchFromAPI('/matches?upcoming=true');
    const data = apiData?.matches || matchesData;
    // Render...
}
```

CSS Variables

```
:root {
    --primary-green: #006400;
    --dark-green: #004d00;
    --light-green: #228b22;
    --accent-gold: #ffd700;
    --black: #000000;
    --white: #ffffff;
    --gray: #6c757d;
    --light-gray: #f8f9fa;
}
```

External Dependencies

Library	Version	CDN
Bootstrap	5.3.0	jsdelivr
Font Awesome	6.4.0	cdnjs

Backend Documentation

File Structure

```
backend/
├── src/
│   ├── config/
│   │   ├── database.js       # MongoDB connection
│   │   └── cloudinary.js     # Cloudinary config
│   ├── models/
│   │   ├── index.js          # Export all models
│   │   ├── User.js           # User model
│   │   ├── Booking.js        # Booking model
│   │   ├── Match.js          # Match model
│   │   ├── Flyer.js          # Flyer model
│   │   ├── Gallery.js        # Gallery model
│   │   ├── Pricing.js        # Pricing model
│   │   └── Settings.js      # Settings model
│   ├── controllers/
│   │   ├── authController.js     # Auth logic
│   │   ├── bookingController.js  # Booking CRUD
│   │   ├── matchController.js    # Match CRUD
│   │   ├── flyerController.js    # Flyer CRUD
│   │   ├── galleryController.js  # Gallery CRUD
│   │   ├── pricingController.js  # Pricing CRUD
│   │   ├── settingsController.js # Settings CRUD
│   │   └── adminController.js    # Admin stats
│   ├── routes/
│   │   ├── index.js              # Route aggregator
│   │   ├── authRoutes.js         # /api/auth
│   │   ├── bookingRoutes.js      # /api/bookings
│   │   ├── matchRoutes.js        # /api/matches
│   │   ├── flyerRoutes.js        # /api/flyers
│   │   ├── galleryRoutes.js      # /api/gallery
│   │   ├── pricingRoutes.js      # /api/pricing
│   │   ├── settingsRoutes.js     # /api/settings
│   │   └── adminRoutes.js        # /api/admin
│   ├── middleware/
│   │   ├── auth.js              # JWT auth
│   │   ├── validate.js           # Validation
│   │   ├── validators.js        # Input validators
│   │   ├── errorHandler.js      # Error handling
│   │   └── rateLimiter.js       # Rate limiting
│   ├── services/
│   │   ├── emailService.js      # Nodemailer
│   │   └── whatsappService.js   # Twilio
│   ├── utils/
│   │   ├── seed.js              # Database seeder
│   │   ├── helpers.js           # Utility functions
│   │   └── catchAsync.js        # Async wrapper
│   └── server.js                # Entry point
├── admin/
```

```
│   ├── login.html
│   ├── dashboard.html
│   ├── bookings.html
│   ├── matches.html
│   ├── flyers.html
│   ├── gallery.html
│   ├── pricing.html
│   ├── settings.html
│   └── js/
│       └── admin.js             # Admin API layer
├── package.json
├── .env.example
└── README.md
```

Server Configuration (server.js)

```
require('dotenv').config();
const express = require('express');
const cors = require('cors');
const helmet = require('helmet');
const connectDB = require('./config/database');
const routes = require('./routes');
const { errorHandler, notFound } = require('./middleware/errorHandler');
const { publicLimiter } = require('./middleware/rateLimiter');

const app = express();

// Security
app.use(helmet());
app.use(cors({ origin: process.env.FRONTEND_URL }));

// Body parsing
app.use(express.json({ limit: '10mb' }));
app.use(express.urlencoded({ extended: true }));

// Rate limiting
app.use('/api', publicLimiter, routes);

// Error handling
app.use(notFound);
app.use(errorHandler);

const PORT = process.env.PORT || 5000;
app.listen(PORT, () => console.log(`Server running on ${PORT}`));
```

Security Features

1. **Helmet.js** - Security headers
2. **CORS** - Cross-origin request control
3. **Rate Limiting** - Prevents abuse

4. **JWT Authentication** - Secure token-based auth
5. **Input Validation** - Prevents injection attacks
6. **Password Hashing** - bcrypt with 12 rounds

Database Documentation

MongoDB Connection

```
// src/config/database.js
const connectDB = async () => {
    try {
        const conn = await mongoose.connect(process.env.MONGODB_URI, {
            maxPoolSize: 10,
            serverSelectionTimeoutMS: 5000,
            socketTimeoutMS: 45000,
        });
        console.log(`MongoDB Connected: ${conn.connection.host}`);
    } catch (error) {
        process.exit(1);
    }
};
```

Data Models

1. User Model

```
{
    name: String,          // Required, max 100 chars
    email: String,         // Required, unique, valid email
    password: String,      // Required, hashed, min 6 chars
    role: String,          // 'admin' | 'user', default: 'user'
    phone: String,         // Optional
    isActive: Boolean,     // default: true
    lastLogin: Date,
    createdAt: Date,
    updatedAt: Date
}
```

2. Booking Model

```
{
    customerName: String,      // Required, max 100 chars
    customerPhone: String,     // Required
    customerEmail: String,     // Optional, valid email
    bookingType: String,      // Required, 'hourly' | 'match' | 'tournament'
    preferredDate: Date,       // Required, future dates
```

```
    preferredTime: String,     // Required
    duration: Number,          // Required, min 1 hour
    numberOfPlayers: Number,   // Required, min 1
    eventType: String,        // Optional
    teamName: String,         // Optional
    opponentTeam: String,     // Optional
    status: String,           // 'pending' | 'confirmed' | 'cancelled' |
'completed'
    totalPrice: Number,       // Default: 0
    notes: String,           // Max 500 chars
    adminNotes: String,
    confirmedBy: ObjectId,   // Ref: User
    confirmedAt: Date,
    cancellationReason: String,
    createdAt: Date,
    updatedAt: Date
}
```

3. Match Model

```
{
    teamA: String,           // Required
    teamB: String,           // Required
    date: Date,              // Required
    time: String,            // Required
    tournament: String,      // Required
    venue: String,          // Default: 'DF Sports Arena'
    status: String,         // 'upcoming' | 'live' | 'completed' | 'cancelled'
    scoreA: Number,
    scoreB: Number,
    bookingId: ObjectId,    // Ref: Booking
    isHomeMatch: Boolean,   // Default: true
    matchDayFlyer: String,
    result: String,         // 'teamA_win' | 'teamB_win' | 'draw'
    createdAt: Date,
    updatedAt: Date
}
```

4. Flyer Model

```
{
    title: String,           // Required, max 200 chars
    imageUrl: String,        // Required
    imagePublicId: String,   // Cloudinary ID
    date: Date,              // Required
    time: String,            // Required
    description: String,    // Max 500 chars
    location: String,        // Default: 'DF Sports Arena, Adjumani'
```

```
    isActive: Boolean,       // Default: true
    isFeatured: Boolean,     // Default: false
    createdBy: ObjectId,     // Ref: User
    createdAt: Date,
    updatedAt: Date
}
```

5. Gallery Model

```
{
    imageUrl: String,        // Required
    imagePublicId: String,   // Cloudinary ID
    altText: String,         // Required, max 100 chars
    category: String,        // 'matches' | 'training' | 'events' | 'facilities' |
'general'
    isActive: Boolean,       // Default: true
    isFeatured: Boolean,     // Default: false
    uploadedBy: ObjectId,  // Ref: User
    createdAt: Date,
    updatedAt: Date
}
```

6. Pricing Model

```
{
    type: String,            // Required, unique, 'hourly' | 'match' | 'tournament'
    name: String,            // Required
    price: Number,           // Required, min 0
    currency: String,        // Default: 'UGX'
    description: String,
    features: [String],
    duration: String,
    isActive: Boolean,       // Default: true
    isPopular: Boolean,      // Default: false
    updatedBy: ObjectId,     // Ref: User
    createdAt: Date,
    updatedAt: Date
}
```

7. Settings Model

```
{
    arenaName: String,       // Default: 'DF Sports Arena'
    tagline: String,
    phone: String,
    whatsapp: String,
```

```
    email: String,
    address: {
        street: String,
        district: String,
        country: String
    },
    location: {
        latitude: Number,
        longitude: Number
    },
    mapEmbedUrl: String,
    operatingHours: {
        open: String,       // Default: '06:00'
        close: String,      // Default: '22:00'
        days: [String]
    },
    socialLinks: {
        facebook: String,
        instagram: String,
        twitter: String,
        youtube: String
    },
    bookingSettings: {
        advanceBookingDays: Number,
        minBookingHours: Number,
        maxBookingHours: Number,
        requireDeposit: Boolean
    },
    notifications: {
        emailNotifications: Boolean,
        whatsappNotifications: Boolean
    },
    createdAt: Date,
    updatedAt: Date
}
```

Database Indexes

```
// Booking indexes
bookingSchema.index({ preferredDate: 1, preferredTime: 1 });
bookingSchema.index({ status: 1 });
bookingSchema.index({ customerPhone: 1 });
bookingSchema.index({ createdAt: -1 });

// Match indexes
matchSchema.index({ date: 1, status: 1 });
matchSchema.index({ tournament: 1 });

// Flyer indexes
flyerSchema.index({ date: 1, isActive: 1 });
```

```
// Gallery indexes
galleryschema.index({ category: 1, isActive: 1 });
```

API Reference

Base URL

```
http://localhost:5000/api
```

Authentication

All protected routes require:

```
Authorization: Bearer <token>
```

Public Endpoints

Health Check

```
GET /api/health
```

Response:

```
{
    "success": true,
    "message": "DF Sports Arena API is running",
    "timestamp": "2026-04-01T12:00:00.000Z"
}
```

Get Upcoming Matches

```
GET /api/matches?upcoming=true&limit=4
```

Response:

```
{
    "success": true,
    "data": {
```

```
        "matches": [
            {
                "_id": "...",
                "teamA": "DF Arena FC",
                "teamB": "Moyo United",
                "date": "2026-04-05T00:00:00.000Z",
                "time": "3:00 PM",
                "tournament": "Adjumani District League",
                "status": "upcoming"
            }
        ],
        "pagination": { "page": 1, "limit": 4, "total": 4, "pages": 1 }
    }
}
```


Get Flyers

```
GET /api/flyers?limit=10
```

Get Gallery

```
GET /api/gallery?limit=20
```

Check Availability

```
GET /api/bookings/availability?date=2026-04-15
```

Response:


```
{
    "success": true,
    "data": {
        "date": "2026-04-15",
        "bookedSlots": [{ "startTime": "10:00", "duration": 2 }],
        "isAvailable": false
    }
}
```


Submit Booking

```
POST /api/bookings
Content-Type: application/json

{
    "customerName": "John Doe",
    "customerPhone": "+256 776 123 456",
    "customerEmail": "john@example.com",
    "bookingType": "match",
    "preferredDate": "2026-04-15",
    "preferredTime": "3:00 PM",
    "duration": 2,
    "numberOfPlayers": 22,
    "teamName": "My Team",
    "opponentTeam": "Opponent Team",
    "notes": "Need extra facilities"
}
```

Response:

```
{
    "success": true,
    "message": "Booking submitted successfully. We will contact you shortly.",
    "data": {
        "booking": {
            "_id": "64a1b2c3d4e5f6789012345",
            "status": "pending",
            ...
        }
    }
}
```

Get Public Settings

```
GET /api/settings/public
```

Protected Endpoints (Admin)

Login

```
POST /api/auth/login
Content-Type: application/json

{
    "email": "admin@dfsportsarena.com",
```

```
    "password": "Admin@123"
}
```

Response:

```
{
    "success": true,
    "message": "Login successful",
    "data": {
        "user": { "_id": "...", "name": "Admin User", "role": "admin" },
        "token": "eyJhbGciOiJIUzI1NiIs..."
    }
}
```

Dashboard Stats

```
GET /api/admin/dashboard
Authorization: Bearer <token>
```

Response:

```
{
    "success": true,
    "data": {
        "overview": {
            "totalBookings": 45,
            "pendingBookings": 5,
            "confirmedBookings": 38,
            "totalRevenue": 4500000
        },
        "recentBookings": [...],
        "bookingStats": [...]
    }
}
```

Create Match

```
POST /api/matches
Authorization: Bearer <token>
Content-Type: application/json

{
    "teamA": "DF Arena FC",
    "teamB": "New Team FC",
```

```
    "date": "2026-05-01",
    "time": "4:00 PM",
    "tournament": "Friendly Match"
}
```

Update Booking

```
PUT /api/bookings/:id
Authorization: Bearer <token>
Content-Type: application/json

{
    "status": "confirmed",
    "totalPrice": 150000,
    "adminNotes": "Confirmed for full match"
}
```

Error Responses

```
{
    "success": false,
    "message": "Error description"
}
```

Common HTTP status codes:

Code	Meaning
200	Success
201	Created
400	Bad Request / Validation Error
401	Unauthorized
403	Forbidden
404	Not Found
429	Too Many Requests
500	Server Error

Admin Panel Guide

Access

URL: `backend/admin/login.html`

Login Credentials

```
Email: admin@dfsportsarena.com
Password: Admin@123
```

Dashboard

- Total bookings count
- Pending bookings count
- Total revenue
- Upcoming matches count
- Recent bookings table
- Booking statistics charts

Bookings Management

- Filter by status (pending/confirmed/cancelled/completed)
- View booking details
- Confirm pending bookings
- Cancel bookings
- Pagination support

Matches Management

- Add new match
- Edit existing match
- Delete match
- View match details

Flyers Management

- Add new flyer
- Upload flyer image
- Edit flyer details
- Delete flyer

Gallery Management

- Upload images
- Set alt text
- Categorize images
- Delete images

Pricing Management

- View all pricing tiers

- Edit price
- Update description

Settings Management

- Arena name and tagline
- Contact information
- Operating hours
- Social media links
- Map embed URL

Installation Guide

Prerequisites

- Node.js 18+
- MongoDB Atlas account
- npm or yarn

Step 1: Clone Repository

```
git clone <repository-url>
cd df-sports-arena
```

Step 2: Backend Setup

```
cd backend
npm install
```

Step 3: Environment Variables

```
cp .env.example .env
```

Edit `.env`:

```
NODE_ENV=development
PORT=5000
MONGODB_URI=mongodb+srv://<username>:<password>@cluster.mongodb.net/dfsportsarena
JWT_SECRET=your-super-secret-key-change-this
JWT_EXPIRES_IN=7d
```

Step 4: Seed Database

```
npm run seed
```

Step 5: Start Server

```
# Development
npm run dev

# Production
npm start
```

Step 6: Frontend

Open `index.html` in browser, or:

```
npx serve .
```

Configuration

Environment Variables

Variable	Required	Description
`PORT`	No	Server port (default: 5000)
`MONGODB_URI`	Yes	MongoDB connection string
`JWT_SECRET`	Yes	JWT signing secret
`JWT_EXPIRES_IN`	No	Token expiry (default: 7d)
`CLOUDINARY_*`	No	Cloudinary credentials
`TWILIO_*`	No	Twilio WhatsApp credentials
`SMTP_*`	No	Email credentials
`ARENA_PHONE`	No	Arena phone number
`ARENA_EMAIL`	No	Arena email
`FRONTEND_URL`	No	Frontend URL for CORS

Getting MongoDB URI

1. Create account at mongodb.com/cloud/atlas
2. Create free cluster
3. Click Connect > Connect your application

4. Copy connection string
5. Replace `<password>` with your user password

Cloudinary Setup (Optional)

1. Sign up at cloudinary.com
2. Copy credentials from Dashboard
3. Add to `.env`

Twilio Setup (Optional)

1. Sign up at twilio.com
2. Get Account SID and Auth Token
3. Enable WhatsApp sandbox or get Business API
4. Add to `.env`

Troubleshooting

MongoDB Connection Failed

```
Error: MongoNetworkError: Failed to connect to server
```

Solution:

- Check MongoDB URI is correct
- Whitelist your IP in Atlas
- Check internet connection

CORS Error

```
Access-Control-Allow-Origin blocked
```

Solution:

- Set `FRONTEND_URL` in `.env`
- Ensure frontend URL matches exactly

JWT Token Invalid

```
Error: JsonWebTokenError: invalid token
```

Solution:

- Clear browser localStorage

- Login again to get new token
- Check JWT_SECRET matches

Image Upload Failed

```
Error: File too large
```

Solution:

- Max file size is 5MB
- Supported formats: jpg, png, webp

WhatsApp Not Sending

```
Error: Twilio not configured
```

Solution:

- Add Twilio credentials to `.env`
- Ensure WhatsApp sandbox is configured

Frontend API Not Loading

The frontend works without backend - it uses static data as fallback.

File Locations Summary

File	Location	Purpose
Server	`backend/src/server.js`	Express server
Models	`backend/src/models/`	MongoDB schemas
Routes	`backend/src/routes/`	API endpoints
Controllers	`backend/src/controllers/`	Business logic
Admin JS	`backend/admin/js/admin.js`	Admin API calls
Frontend JS	`js/script.js`	Main frontend logic
Styles	`css/style.css`	All CSS styles
Database Seed	`backend/src/utils/seed.js`	Sample data

Support

For questions or issues:

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